

Math 4441: Probability and Statistics

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Chapter 3: Well-Known Random Variables and Distributions

(Part 2 – Bivariate and Multivariate Well-known RVs)

Lectures 19

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References:

1. Goodman and Yates – Introduction to Probability and Stochastic Processes, 3rd Edition
2. Scott Miller and Donald Childers – Probability and Random Processes, 2nd Edition
3. Bertsekas and Tsitsiklis – Introduction to Probability, 2nd Edition

Well-Known bivariate and multivariate Random Variables

1. Multinomial Random Variables
2. Multivariate Hypergeometric Random Variables
3. Categorical Random Variables
4. Bivariate Gaussian/Normal Random Variables

Example: Consider an urn contains balls of three colors – 10 red balls, 8 blue balls and 7 green balls. Suppose you pick 5 balls from the urn – (i) without replacement and (ii) with replacement.

Find the probability that you will pick exactly 2 red balls, 2 blue balls and 1 green ball.

i) Without replacement

Multivariate Hypergeometric Random Variables

Definition 5.1. (*Multivariate hypergeometric distribution*) Suppose an urn contains items of r different types, where the number of the i -th type is n_i and the number of total items is $n_1 + n_2 + \cdots + n_r$. Now, n items are drawn from the urn at random. Let the random variable X_i , where $i = 1, 2, \dots, r$, represent the number of items drawn of the i -th type. Then, the multivariate joint probability mass function of $X_1, X_2, \dots, X_r$ is given by

$$P_{X_1, \dots, X_r}(x_1, \dots, x_r) = \frac{\binom{n_1}{x_1} \binom{n_2}{x_2} \cdots \binom{n_r}{x_r}}{\binom{n_1 + n_2 + \cdots + n_r}{n}},$$

where $x_1 + x_2 + \cdots + x_r = n$. The above MJPMF is known as the multivariate hypergeometric distribution.

ii) With Replacement



Multinomial Distribution: If $X_1, X_2, \dots, X_k$ are a set of discrete random variables, then they have the multinomial distribution with parameters n and $p_1, p_2, \dots, p_k$, where $p_1 + p_2 + \dots + p_k = 1$

$$P_{X_1, X_2, \dots, X_k}(x_1, x_2, \dots, x_k) = \begin{cases} \binom{n}{x_1, x_2, \dots, x_k} p^{x_1} \dots p^{x_k}, & x_1 + \dots + x_k = n \\ 0, & \text{otherwise} \end{cases}$$

$$\binom{n}{x_1, \dots, x_k} = \frac{n!}{x_1! x_2! \dots x_k!}.$$

Approximation: If in a multivariate hypergeometric distribution, the number of items available is very large compared to the number of items selected, then the multivariate hypergeometric distribution can be approximated by the multinomial distribution.

Example

Attendance at a Baseball Game. Suppose that 23 percent of the people attending a certain baseball game live within 10 miles of the stadium, 59 percent live between 10 and 50 miles from the stadium, and 18 percent live more than 50 miles from the stadium. Suppose also that 20 people are selected at random from the crowd attending the game. We shall determine the probability that seven of the people selected live within 10 miles of the stadium, eight of them live between 10 and 50 miles from the stadium, and five of them live more than 50 miles from the stadium.

$$\frac{20!}{7!8!5!}(0.23)^7(0.59)^8(0.18)^5 = 0.0094.$$

Example: Consider an urn contains balls of three colors – 10 red balls, 8 blue balls and 7 green balls. Suppose you pick 5 balls from the urn with replacement. Find the probability that you will pick exactly 2 red balls and at least 2 blue balls.

$$P(X_1 = 3, X_2 \geq 2) = P(X_1 = 3, X_2 = 2, X_3 = 2) \\ + P(X_1 = 3, X_2 = 3, X_3 = 1) + P(X_1 = 3, X_2 = 4, X_3 = 0)$$

Categorical Distribution: If there are k possible outcomes in an experiment and any of these outcomes is considered as a success and others as failures ($x_k = 1$, if category k is observed and 0 otherwise), then this becomes a generalization of the Bernoulli distribution

Bivariate Gaussian Distribution

Random variables X and Y are jointly Gaussian if their joint PDF is given by

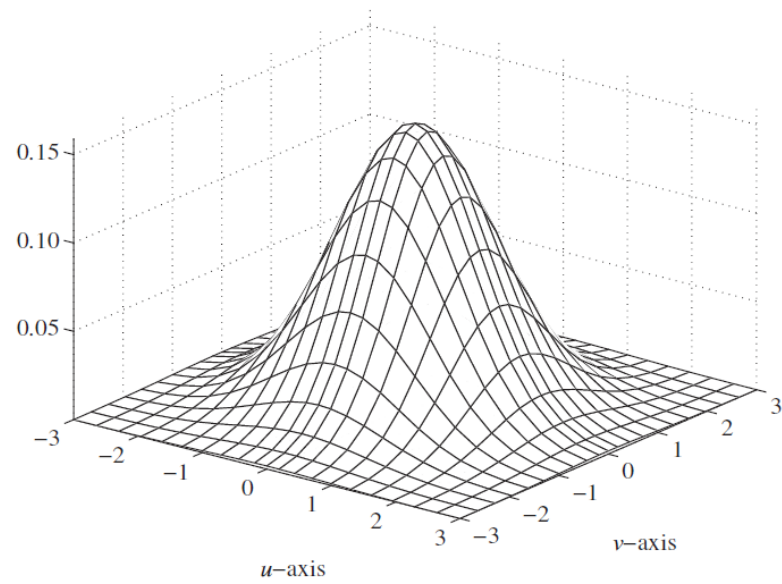
$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho_{XY}^2}} e^{-\frac{\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho_{XY}\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2}{2(1-\rho_{XY}^2)}}$$

where

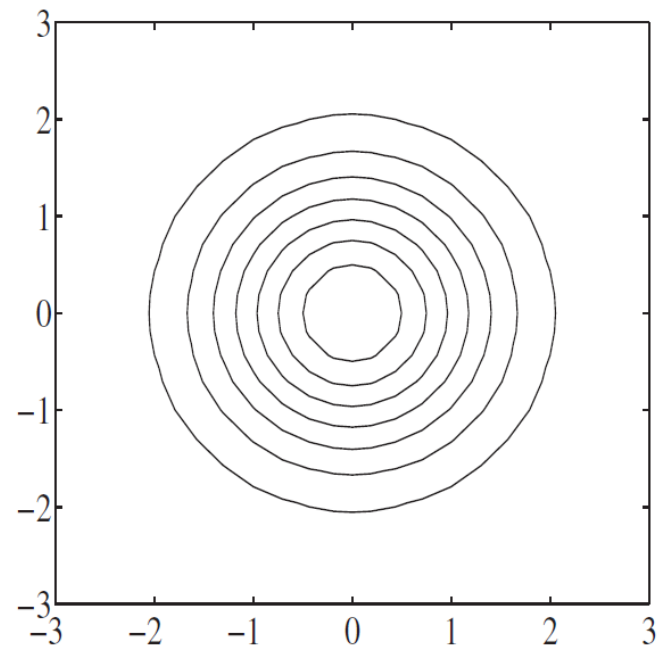
μ_X and μ_Y are means of X and Y

σ_X and σ_Y are standard deviations of X and Y

ρ_{XY} is the covariance of X and Y .



The bivariate Gaussian surface $\psi_0(z_1, z_2)$ for independent standard Gaussian



Level curves of the bivariate Gaussian pdf $\psi_0(z_1, z_2)$ for independent standard

The marginal PDF of X is given by

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma_X^2}} e^{-\frac{(x-\mu_X)^2}{2\sigma_X^2}}$$

The marginal PDF of Y is given by

$$f_Y(y) = \frac{1}{\sqrt{2\pi\sigma_Y^2}} e^{-\frac{(y-\mu_Y)^2}{2\sigma_Y^2}}$$